

1. Calculus Cliffs — Differentiation
 a. Exponential/logarithmic derivative

Example 1

Differentiate

a) e^{7x}

b) e^{5x-4}

c) e^{x^3} .

a) $\frac{d}{dx} e^{7x} = 7 e^{7x}$

Substitute for $k = 7$ in $k e^{kx}$.

b) $\frac{d}{dx} e^{5x-4} = 5 e^{5x-4}$

Use $\frac{dy}{dx} = f'(x) e^{f(x)}$ where $f'(x) = 5$.

c) $\frac{d}{dx} e^{x^3} = 3x^2 e^{x^3}$

Use $\frac{dy}{dx} = f'(x) e^{f(x)}$ where $f'(x) = 3x^2$.

Example 2

Find $\frac{dy}{dx}$ when

a) $y = 4e^{2x}$

b) $y = -2e^{\sqrt{x}}$

c) $y = 3e^{\frac{1}{x}}$.

a) $\frac{dy}{dx} = 4e^{2x} (2) = 8e^{2x}$

Write down $4e^{2x}$ and then multiply by $\frac{d}{dx} 2x$.

b) $\frac{dy}{dx} = -2e^{\sqrt{x}} \left(\frac{1}{2} x^{-\frac{1}{2}} \right)$
 $= -\frac{1}{\sqrt{x}} e^{\sqrt{x}}$

Write down $-2e^{\sqrt{x}}$ and then multiply by $\frac{d}{dx} x^{\frac{1}{2}}$.

c) $\frac{dy}{dx} = 3e^{\frac{1}{x}} (-1x^{-2})$
 $= -\frac{3e^{\frac{1}{x}}}{x^2}$

Write down $3e^{\frac{1}{x}}$ and then multiply by $\frac{d}{dx} x^{-1}$.

Example 3

Find $\frac{dy}{dx}$ when **a)** $y = \ln 5x$ **b)** $y = \ln(3x^2 - 2)$.

a) Let $u = 5x$, then $\frac{du}{dx} = 5$

$$y = \ln u, \text{ so } \frac{dy}{du} = \frac{1}{u}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u} \times 5 = \frac{5}{u}$$

$$\text{Thus } \frac{dy}{dx} = \frac{5}{5x} = \frac{1}{x}$$

Use the chain rule.

Substitute $u = 5x$ to have your answer in terms of x .

Note: Alternatively, we could say $y = \ln 5x = \ln 5 + \ln x$, so $\frac{dy}{dx} = 0 + \frac{1}{x} = \frac{1}{x}$ (since $\ln 5$ is a constant).

b) Let $u = 3x^2 - 2$, then $\frac{du}{dx} = 6x$

$$y = \ln u, \text{ so } \frac{dy}{du} = \frac{1}{u}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \frac{1}{u} \times 6x = \frac{6x}{u}$$

$$\text{Thus } \frac{dy}{dx} = \frac{6x}{3x^2 - 2}$$

Use the chain rule.

Eliminate u to present your answer in terms of x .

If $y = \ln[f(x)]$, then $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$.

Example 4

Differentiate

a) $\ln(9x + 2)$ **b)** $\ln(e^x - 3x)$ **c)** $4 \ln 8x$ **d)** $\ln(5x - 1)^7$.

a) $\frac{dy}{dx} = \frac{f'(x)}{f(x)} = \frac{9}{9x + 2}$

$f(x) = (9x + 2), f'(x) = 9$

b) $\frac{dy}{dx} = \frac{e^x - 3}{e^x - 3x}$

$f(x) = (e^x - 3x), f'(x) = e^x - 3$

c) $\frac{dy}{dx} = 4 \times \frac{f'(x)}{f(x)} = 4 \times \frac{8}{8x} = \frac{4}{x}$

$f(x) = 8x, f'(x) = 8$

d) $\ln(5x - 1)^7 = 7 \ln(5x - 1)$

$$\frac{dy}{dx} = 7 \times \frac{f'(x)}{f(x)} = 7 \times \frac{5}{5x - 1} = \frac{35}{5x - 1}$$

It is best to use laws of logarithms (covered in Chapter 2) to first simplify the expression.

1. Differentiate

- | | | |
|----------------------|--------------------|-----------------------|
| a) e^{2x} | b) e^{-5x} | c) e^{3x+9} |
| d) e^{8-x} | e) $-7e^x$ | f) $5e^{x^2}$ |
| g) $3e^{2x-1}$ | h) $-4e^{6+x^2}$ | i) e^{5x^2} |
| j) $e^{\frac{3}{x}}$ | k) $6e^{\sqrt{x}}$ | l) $\frac{1}{e^{2x}}$ |

2. Find $\frac{dy}{dx}$ when

- | | | |
|--------------------------|----------------------------|--------------------------------|
| a) $y = e^{-2x}$ | b) $y = e^{5+x}$ | c) $y = 9e^{3x}$ |
| d) $y = -8e^{\sqrt{x}}$ | e) $y = 3e^x - 2e^{4x}$ | f) $y = \frac{1}{2}e^{-8x}$ |
| g) $y = e^{4x} + e^{-x}$ | h) $y = 6e^{-x} - 4e^{3x}$ | i) $y = (e^x + 2)(e^x - 1)$ |
| j) $y = (2e^{3x} - 1)^2$ | k) $y = \frac{1+e^x}{e^x}$ | l) $y = e^{-4x+2} + 7e^{5-2x}$ |

Hint for part (i): First expand the brackets.

3. Find the gradient of the tangent to the curve $y = 5e^{2x}$ at the point (0, 5).

4. Find the exact value of the gradient of the tangent to the curve $y = e^x - 6\sqrt{x}$ when $x = 1$.

1. Differentiate

- | | | | |
|------------------------|-------------------|------------------------|-----------------------------|
| a) $\ln 4x$ | b) $\ln(1 - x^2)$ | c) $\ln(3 + 5x)$ | d) $\ln(x^3 + 2)$ |
| e) $\ln(e^x - 7)$ | f) $\ln\sqrt{x}$ | g) $\ln(6x - 3)^4$ | h) $\ln(4x^2 + 2x)$ |
| i) $\ln \frac{1}{x^2}$ | j) $3 \ln 4x$ | k) $2 \ln \frac{1}{x}$ | l) $6 \ln \frac{9x - 2}{3}$ |

2. Find $\frac{dy}{dx}$ when

- | | | | |
|--|-------------------------------|-------------------------|--------------------------------|
| a) $y = 8 \ln 3x$ | b) $y = \ln 2x^3$ | c) $y = \ln(2 - 4x)$ | d) $y = 6 \ln \frac{x}{2}$ |
| e) $y = \ln\sqrt{2x+1}$ | f) $y = \ln(5x)^{10}$ | g) $y = \ln(3e^x - 2x)$ | h) $y = 9 \ln(x^2 + 3x)^2$ |
| i) $y = 5 \ln\left(7 - \frac{2}{x}\right)$ | j) $y = 4 \ln\sqrt{x^2 - 8x}$ | k) $y = \ln e^{5x}$ | l) $y = \ln \frac{7}{(x+3)^6}$ |

3. Find the gradient of the tangent to the curve $y = 4 + \ln x$ at the point (1, 4).

4. Find the gradient of the tangent to the curve $y = \ln(x^2 + 1)$ when $x = 3$.

5. Find the exact coordinates of the point on the curve $y = \ln 3x$ where the gradient is $\frac{1}{2}$.

b. Product rule

Example 5

Find $\frac{dy}{dx}$ when $y = e^{2x}(x^3 - 3)$.

Let $y = uv$ where $u = e^{2x}$ and $v = (x^3 - 3)$.

y is a product of two functions.

$$\frac{du}{dx} = 2e^{2x} \text{ and } \frac{dv}{dx} = 3x^2$$

Differentiate u and differentiate v .

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Use the product rule.

$$\frac{dy}{dx} = (e^{2x})(3x^2) + (x^3 - 3)(2e^{2x})$$

Substitute for each of the four unknowns.

$$= e^{2x}(2x^3 + 3x^2 - 6)$$

Factorise by taking out the common e^{2x} .

Example 6

Find $\frac{dy}{dx}$ when $y = (2x + 1)(x - 5)^4$.

Let $y = uv$ where $u = (2x + 1)$ and $v = (x - 5)^4$.

y is a product of two functions.

$$\frac{du}{dx} = 2 \text{ and } \frac{dv}{dx} = 4(x - 5)^3(1)$$

Differentiate u and differentiate v .

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Use the product rule.

$$\frac{dy}{dx} = (2x + 1)[4(x - 5)^3] + [(x - 5)^4](2)$$

Substitute for each of the four unknowns.

$$= 2(x - 5)^3(4x + 2 + x - 5)$$

$$= 2(x - 5)^3(5x - 3)$$

Factorise by taking out the common factor $2(x - 5)^3$.

1. Use the product rule to find $\frac{dy}{dx}$ when

a) $y = x e^x$

b) $y = (4x - 1)(x^2 - 5)$

c) $y = x^2 \sqrt{x^2 + 1}$

d) $y = \sqrt{x}(\ln x)$

e) $y = x^2(2x + 5)^7$

f) $y = \sqrt{x}(x - 3)^2$

g) $y = e^{2x}(4x + 1)$

h) $y = x(x + 3)^4$

i) $y = x^2 \ln x$

j) $y = x^2(x + 3)^3$

k) $y = e^x(2x - 1)^4$

l) $y = \frac{x}{x + 1}$

m) $y = 6e^{\sqrt{x}}(x + 4)$

n) $y = 4x^3 e^{2x}$

o) $y = 2e^x \ln 2x$

p) $y = \frac{x - 2}{x + 2}$

q) $y = \frac{2x + 3}{1 - 5x}$

r) $y = (x^2 + 1) \ln(x^2 + 1).$

Hint for
 $y = x(x + 3)^3$
the product rule

2. Find the gradient of the tangent to the curve $y = x^2 e^{-x}$ at the point $\left(1, \frac{1}{e}\right)$.

3. Find the equation of the tangent to the curve $y = x(1 + x^2)^3$ when $x = 2$.

4. Show that $\frac{dy}{dx} = \frac{2x^2 + 4x - 1}{\sqrt{(x^2 - 1)}}$ when $y = (x + 4)\sqrt{(x^2 - 1)}$.

c. Quotient rule

Example 8

Find $\frac{dy}{dx}$ when $y = \frac{x^3}{x^2 + 1}$.

Let $y = \frac{u}{v}$ where $u = x^3$ and $v = x^2 + 1$.

$$\frac{du}{dx} = 3x^2 \text{ and } \frac{dv}{dx} = 2x$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{dy}{dx} = \frac{(x^2 + 1)(3x^2) - (x^3)(2x)}{(x^2 + 1)^2}$$

$$= \frac{3x^4 + 3x^2 - 2x^4}{(x^2 + 1)^2}$$

$$= \frac{x^2(x^2 + 3)}{(x^2 + 1)^2}$$

y is a quotient of two functions.

Differentiate u and differentiate v .

Use the quotient rule.

Substitute for each of the four unknowns.

Factorise by taking out the common x^2 .

Example 9

Find $\frac{dy}{dx}$ when $y = \frac{4x + 3}{\sqrt{2x - 1}}$.

Let $y = \frac{u}{v}$ where $u = 4x + 3$ and $v = \sqrt{2x - 1}$.

$$\frac{du}{dx} = 4$$

$$\frac{dv}{dx} = \frac{1}{2}(2x - 1)^{-\frac{1}{2}}(2) = (2x - 1)^{-\frac{1}{2}}$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$\frac{dy}{dx} = \frac{(2x - 1)^{\frac{1}{2}}(4) - (4x + 3)(2x - 1)^{-\frac{1}{2}}}{2x - 1}$$

$$= \frac{(2x - 1)^{-\frac{1}{2}}[4(2x - 1) - (4x + 3)]}{2x - 1}$$

$$= \frac{4x - 7}{(2x - 1)^{\frac{3}{2}}}$$

Differentiate u and differentiate v .

Use the quotient rule.

Substitute for each of the four unknowns.

Note: $(2x - 1)^{-\frac{1}{2}} \times (2x - 1)^1 = (2x - 1)^{\frac{1}{2}}$

$$\frac{(2x - 1)^{-\frac{1}{2}}}{(2x - 1)^1} = \frac{1}{(2x - 1)^{\frac{3}{2}}}$$

1. Use the quotient rule to find $\frac{dy}{dx}$ when

a) $y = \frac{2x}{x+3}$

b) $y = \frac{e^{2x}}{x}$

c) $y = \frac{x^2}{x+4}$

d) $y = \frac{x}{1+x^2}$

e) $y = \frac{3x}{2-x}$

f) $y = \frac{3x^2}{2e^x - x}$

g) $y = \frac{3x-4}{2x+1}$

h) $y = \frac{3x+5}{(x+1)^2}$

i) $y = \frac{6x^2}{2-x}$

j) $y = \frac{4x}{(1-x)^3}$

k) $y = \frac{x^2+1}{x-2}$

l) $y = \frac{5x^2}{\ln x}$

m) $y = \frac{x^3}{\sqrt{1-2x^2}}$

n) $y = \frac{3x+x^4}{2x^2+1}$

o) $y = \frac{x}{1+\sqrt{x}}$

p) $y = \frac{x^2+6}{2x-7}$

q) $y = \frac{\sqrt{(x+1)^5}}{x}$

r) $y = \frac{1-2x^2}{\sqrt{1+2x^2}}$

2. If $y = \frac{x^2}{2x-3}$ find the values of x at the points where $\frac{dy}{dx} = 0$.

3. Find the equation of the normal to the curve $y = \frac{3x^2}{2x-1}$ at the point $(2, 4)$.

4. If $y = \frac{x}{x-1}$, find $\frac{d^2y}{dx^2}$.

5. Show that the coordinates of the turning point on the curve $y = \frac{x+3}{\sqrt{1+x^2}}$ are $\left(\frac{1}{3}, \sqrt{10}\right)$.

6. Show that if $y = \frac{7\ln x - x^3}{e^{3x}}$ then $\frac{dy}{dx} = \frac{7}{e^3}$ when $x = 1$.

d. Sin/cos/tan derivative

Example 10

Find $\frac{dy}{dx}$ when

a) $y = \cos 2x$ b) $y = 5 \sin 3x^2$ c) $y = 7 \tan \left(4x + \frac{\pi}{2} \right)$.

a) Let $u = 2x \Rightarrow \frac{du}{dx} = 2$

$$y = \cos u \Rightarrow \frac{dy}{du} = -\sin u = -\sin 2x$$

Substitute $u = 2x$.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Use the chain rule to help you.

$$= (-\sin 2x)(2) = -2 \sin 2x$$

Substitute for each of the two unknowns.

b) Let $u = 3x^2 \Rightarrow \frac{du}{dx} = 6x$

Substitute $u = 3x^2$ and use the chain rule to help you differentiate.

$$y = 5 \sin u \Rightarrow \frac{dy}{du} = 5 \cos u$$

$$\begin{aligned} \frac{dy}{dx} &= (5 \cos u)(6x) \\ &= 30x \cos 3x^2 \end{aligned}$$

Using the chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

c) Let $u = 4x + \frac{\pi}{2} \Rightarrow \frac{du}{dx} = 4$

Substitute $u = 4x + \frac{\pi}{2}$.

$$y = 7 \tan u \Rightarrow \frac{dy}{du} = 7 \sec^2 u$$

$$\frac{dy}{dx} = (7 \sec^2 u) \times 4$$

Using the chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

$$= 28 \sec^2 \left(4x + \frac{\pi}{2} \right)$$

1. Find $\frac{dy}{dx}$ when

a) $y = \sin 5x$

d) $y = 4 \tan^2 x$

g) $y = \cos^3 x$

j) $y = \frac{\cos 2x}{x^2}$

b) $y = 6 \tan \frac{x}{2}$

e) $y = \sin(x^3 - 7)$

h) $y = \sin 2x \cos x$

k) $y = \sin^3 3x \cos x$

c) $y = \cos(x^2 + 2x)$

f) $y = 5 \tan 4x$

i) $y = x^2 \tan 3x$

l) $y = 4 \tan \sqrt{x}$.

e. Implicit derivative

Example 11

Find $\frac{dy}{dx}$ in terms of x and y when $x^2 - 3y^3 = 4y^2$.

$$2x - 9y^2 \frac{dy}{dx} = 8y \frac{dy}{dx}$$



Differentiate each term with respect to x .

$$2x = 8y \frac{dy}{dx} + 9y^2 \frac{dy}{dx}$$



Rearrange terms.

$$\frac{dy}{dx}(8y + 9y^2) = 2x$$



Factorise.

$$\frac{dy}{dx} = \frac{2x}{8y + 9y^2}$$



Rearrange for $\frac{dy}{dx}$.

Example 12

Find the equation of the normal to the curve $5x^2 + 6xy - y^2 = 10$ at the point $(1, 5)$.

$$10x + 6x \frac{dy}{dx} + 6y - 2y \frac{dy}{dx} = 0$$

← Use the product rule to differentiate $6xy$.

$$(6x - 2y) \frac{dy}{dx} = -10x - 6y$$

← Rearrange terms.

$$\text{At } (1, 5) \Rightarrow (6 - 10) \frac{dy}{dx} = -10 - 30$$

← Substitute for $x = 1$ and $y = 5$.

$$\text{So } \frac{dy}{dx} = \frac{-40}{-4} = 10$$

Thus gradient of tangent to curve at $(1, 5) = 10$

$$\text{Gradient of normal to curve at } (1, 5) = -\frac{1}{10}$$

← $m_1 \times m_2 = -1$ for perpendicular lines.

Equation of normal at $(1, 5)$ is

$$y - 5 = -\frac{1}{10}(x - 1)$$

← $(y - y_1) = m(x - x_1)$ is the equation of a straight line.

$$\text{or } -10y + 50 = x - 1$$

← Multiply both sides by -10 .

$$\text{or } 10y = 51 - x$$

Example 13

It is given that $y = \tan^{-1} x$. Find an expression for $\frac{dy}{dx}$.

$$y = \tan^{-1} x$$

$$x = \tan y$$

← Use what you learned in P1.

$$1 = \sec^2 y \frac{dy}{dx}$$

← Differentiate with respect to x implicitly.

$$1 = (1 + \tan^2 y) \frac{dy}{dx}$$

← Use $1 + \tan^2 y \equiv \sec^2 y$.

Rearranging:

$$\frac{1}{1 + \tan^2 y} = \frac{dy}{dx}$$

and substituting $x = \tan y$ gives

$$\frac{dy}{dx} = \frac{1}{1 + x^2}$$

1. Find $\frac{dy}{dx}$ in terms of x and y when

a) $x^2 + 2y - y^2 = 5$

b) $x \cos y = y^2 + x$

c) $y = 5 \tan^{-1} 3x$

d) $x^3 + xy^2 = 5x$

e) $x^2 + y^2 = y$

f) $2xy - 3y = y^2 - 7x$

g) $x \ln y = 1 + x$

h) $y = e^x \tan^{-1} x$

i) $x^2 + y^2 + 2x - 4y + 4 = 0$

j) $x^4 + x^2y^2 - y^4 = 5$

k) $x \sin y - y \sin x = 8$

l) $e^{x+y} = 2x$

m) $y = x \tan^{-1} x$

n) $y + y^3 = x - x^2$

o) $xe^y + 2y = 1$

p) $y = x \ln y$

q) $y = \frac{1}{4} \tan^{-1} 2x$

r) $\sin y + x^2y^3 = 2y + \cos x.$

2. Find the gradients of the tangents to the curve $x^2 + 6y^2 = 10$ when $x = 2$.

3. Find the equation of the tangent to the curve $x^3 + y^3 = 2xy$ at the point $(1, 1)$.

4. Show that $\frac{dy}{dx} = \frac{2y^2 - 2xy^3}{3x^2y^2 - 4xy - 3}$ when $x^2y^3 - 2xy^2 = 3y$.

5. Given that $\ln(xy) = x^2 + y^2$, show that $\frac{dy}{dx} = \frac{y(2x^2 - 1)}{x(1 - 2y^2)}$.

f. Parametric derivative

When differentiating parametric equations involving a parameter t ,

$$\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$$

This is called **parametric differentiation**.

Example 14

a) If $x = 2t^3$ and $y = 5t^2 - 6$, find $\frac{dy}{dx}$ in terms of t .

b) Find the value of $\frac{dy}{dx}$ when $x = -16$.

a) $\frac{dx}{dt} = 6t^2, \frac{dy}{dt} = 10t$

← Differentiate x and y with respect to t .

$$\frac{dy}{dx} = \frac{10t}{6t^2} = \frac{5}{3t}$$

← $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$

b) When $x = -16 \Rightarrow -16 = 2t^3$

← Use $x = -16$ to find the value of t .

$$\Rightarrow t = -2$$

$$\frac{dy}{dx} = \frac{5}{-6} = -\frac{5}{6}$$

← Substitute this value of t into the expression you found in (a).

Example 15

The parametric equations of a curve are given by $x = 2 \cos \theta$, $y = \sin^2 \theta$, $0 \leq \theta \leq 2\pi$.

Find the values of θ when the tangent to the curve is parallel to the x -axis.

$$\frac{dx}{d\theta} = -2 \sin \theta, \frac{dy}{d\theta} = 2 \sin \theta \cos \theta$$

← Differentiate x and y with respect to θ .

$$\frac{dy}{dx} = \frac{2 \sin \theta \cos \theta}{-2 \sin \theta} = -\cos \theta$$

← $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$

When $-\cos \theta = 0$,

← $\frac{dy}{dx} = 0$ when tangent parallel to x -axis.

$$\theta = \frac{\pi}{2} \text{ and } \frac{3\pi}{2}$$

← $0 \leq \theta \leq 2\pi$

1. Find $\frac{dy}{dx}$ for the following parametric equations.

Give your answer in its simplest form.

a) $x = 2t^3, \quad y = 4t^2 + 1$

b) $x = t^3 - 6t, \quad y = t + \frac{2}{t}$

c) $x = 3 \cos t, \quad y = 2 \sin t$

d) $x = 4t^2 - 7, \quad y = 7 + 8t$

e) $x = t - \sin t, \quad y = 2 - \cos t$

f) $x = \frac{1}{1-t}, \quad y = t^2 - 9$

g) $x = t^2 - 3, \quad y = 2 + \sqrt{t}$

h) $x = 3 \cos^3 \theta, \quad y = 3 \sin^3 \theta$

i) $x = t^2 + 3t - 1, \quad y = 3(t - 2)$

j) $x = \cos 2\theta, \quad y = 4 \sin \theta$

k) $x = a \cos \theta, \quad y = b \sin \theta$

l) $x = 2 - t^2, \quad y = 5 + 4t - t^3$

2. The parametric equations of a curve are $x = t + \frac{1}{t}, y = t - \frac{1}{t}$. Show that $\frac{dy}{dx} = \frac{t^2 + 1}{t^2 - 1}$.

3. The parametric equations of a curve are $x = 4 \sin t, y = \cos 2t$.

Show that $\frac{dy}{dx} = -\sin t$.

4. The parametric equations of a curve are $x = 2 + t, y = \frac{1}{t^2} - 6$. Show that $\frac{dy}{dx} = -2t^{-3}$.

5. The parametric equations of a curve are $x = \cos 2\theta, y = 2\theta + \sin 2\theta$. Show that $\frac{dy}{dx} = -\cot \theta$.